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Done

PYMBLE LADIES' COLLEGE

YEAR 12

MATHEMATICS EXTENSION 1

HSC TRIAL EXAMINATION 2002

Time Allowed: 2 hours + 5 mins reading time

INSTRUCTIONS

- All questions should be attempted
- Write your name and your teacher's name on each page
- Start each question on a new page
- DO NOT staple the questions together
- Only approved calculators may be used
- A standard integral sheet is attached
- Marks might be deducted for careless or untidy work
- Hand this question paper in with your answers
- ALL rough working paper must be attached to the back of the last question
- Staple a coloured sheet of paper to the back of each question
- There are seven (7) questions in this paper
- All questions are of equal value

Question 1

Marks

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$

1

(b) The point P (7, -1) divides the interval AB externally in the ratio 3 : 2.
If A is (-2, 5) find the coordinates of B.

2

(c) Solve for x

$$\frac{x+1}{x-2} < 2$$

2

(d) Find the gradient of the tangent to the curve $y = \tan^{-1}(2x)$ at the point where $x = \frac{1}{2}$.

2

(e) Evaluate $\int_0^1 \frac{1}{\sqrt{9-x^2}} dx$

2

(f) On the same number plane, sketch the graphs of

(i) $y = |2x - 1|$ and $y = |x + 1|$

2

(ii) Hence, or otherwise, solve $|2x - 1| \leq |x + 1|$

1

Question 2 (Start a new sheet of paper)

(a) Prove that $\frac{\sin 2\theta}{\sin \theta} - \sec \theta = \frac{\cos 2\theta}{\cos \theta}$ 2

(b) Evaluate $\int_{\frac{1}{2}}^1 4t(2t-1)^3 dt$ by using the substitution $u = 2t - 1$ 4

(c) The angle between the lines $y = 3x$ and $y = mx$ is 45° . Find the value(s) of m . 3

(d) Solve $\tan 2\theta - \cot \theta = 0$ where $0 \leq \theta \leq \pi$ 3

Question 3 (Start a new sheet of paper)

Marks

(a) Evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^3\left(\frac{x}{2}\right) dx$ 2

(b) Use Mathematical Induction to prove that

(i) $4(1^3 + 2^3 + 3^3 + \dots + n^3) = n^2(n+1)^2$, for $n = 1, 2, 3, \dots$ 3

(ii) Hence find the value of $\lim_{n \rightarrow \infty} \left(\frac{1^3 + 2^3 + 3^3 + \dots + n^3}{n^4} \right)$ 1

(c) (i) Express $\sin x + \sqrt{3} \cos x$ in the form $R \sin(x + \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$ 2

(ii) Hence sketch $y = \sin x + \sqrt{3} \cos x$ for $-2\pi \leq x \leq 2\pi$ showing any x and y intercepts. 2

(iii) Find the general solution to $\sin x + \sqrt{3} \cos x = \sqrt{2}$ 2

Question 4 (Start a new sheet of paper)

Marks

(a) α, β and γ are the roots of the equation $x^3 + 2x^2 - 3x + 5 = 0$

(i) State the values of $\alpha + \beta + \gamma$, $\alpha\beta + \alpha\gamma + \beta\gamma$

2

(ii) Find the value of $\alpha^3 + \beta^3 + \gamma^3$

2

(b) If a polynomial $P(x)$ is divided by $(x+1)$ the remainder is 5 and when $P(x)$ is divided by $(2x+1)$ the remainder is 3. Find the remainder when $P(x)$ is divided by $(x+1)(2x+1)$.

3

(c) From a point S the bearings of two points P and Q are found to be 331° T and 011° T respectively. From a point F , 7 km due north of S , the bearings of P and Q are 299° T and 020° T respectively.

(i) Show that $PF = \sin 29^\circ \times \frac{7}{\sin 32^\circ}$

2

(ii) By considering the triangle FPQ , show that if the distance between P and Q is d metres, then

$$d^2 = 49 \left(\frac{\sin^2 29^\circ}{\sin^2 32^\circ} + \frac{\sin^2 11^\circ}{\sin^2 9^\circ} - 2 \frac{\sin 29^\circ \sin 11^\circ \cos 81^\circ}{\sin 32^\circ \sin 9^\circ} \right)$$

3

Question 5 (Start a new sheet of paper)

Marks

(a) Consider the function $f(x) = \frac{x-1}{x^2}$

(i) Show that there is only one stationary point and determine its nature

3

(ii) Determine the point of inflexion.

1

(iii) What happens to $f(x)$ as $x \rightarrow \pm\infty$?

1

(iv) What happens to $f(x)$ as $x \rightarrow 0$?

1

(v) Sketch the curve showing all its essential features. (Use at least half a page.)

2

(b) (i) Prove that $\frac{d}{dx} \left(\frac{1}{2} x^2 \right) = x$

2

(ii) An object moving in a straight line has an acceleration given by $\ddot{x} = x(8 - 3x)$ where x metres is its position relative to a fixed point O .

At $x = 0$, it has a speed of 4 m/s. Find its speed when it is 1 m on the positive side of O .

2

Question 6 (Start a new sheet of paper)

Marks

- (a) A particle is oscillating in simple harmonic motion such that its displacement x metres from the origin is given by the equation $\frac{d^2x}{dt^2} = -16x$ where t is time in seconds.
- (i) Show that $x = a \cos(4t + \alpha)$ is a solution of motion for this particle. (a and α are constants). 1
- (ii) When $t = 0$, $v = 4$ m/s and $x = 5$ m. Show that the amplitude of the oscillation is $\sqrt{26}$ metres. 2
- (iii) What is the maximum speed of the particle? 1
- (b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$. The tangents at P and Q meet at T which is always on the parabola $x^2 = -4ay$.
- (i) Derive the equation of the tangent at P . 2
- (ii) Hence write down the equation of the tangent at Q . 1
- (iii) Show that T is the point $(a(q + p), apq)$. 1
- (iv) Show that $p^2 + q^2 = -6pq$. 1
- (v) Find M , the midpoint of PQ . 1
- (vi) Hence, or otherwise, find the locus of M . 2

Question 7 (Start a new sheet of paper)

Marks

- (a) (i) On the same number plane, sketch the graphs of $y = \cos^{-1} x$ and $y = \sin^{-1}(\frac{x}{2})$. Label the important features. 2
- (ii) Show $y = \cos^{-1} x$ and $y = \sin^{-1}(\frac{x}{2})$ intersect at $x = \frac{2}{\sqrt{5}}$. 2
- (iii) Find the inverse function of $y = \sin^{-1}(\frac{x}{2})$. 1
- (iv) Hence or otherwise find the area bounded by the x -axis and the graphs $y = \cos^{-1} x$ and $y = \sin^{-1}(\frac{x}{2})$ (answer correct to 2 decimal places.) 3
- (b) Wheat is the only crop grown on Sandy's property in outback NSW. Per hectare the amount of water, W , in kilolitres, used during irrigation times is given by
- $$W = Cg^2 + \frac{D}{g}$$
- where g is the amount of grain produced in tonnes per hectare and C and D are positive constants. There is a limited amount of water available for irrigation.
- (i) Show that, for maximum hectares under irrigation, production of grain per hectare, g , is given by
- $$g = \left(\frac{D}{2C}\right)^{\frac{1}{3}}$$
- (ii) Show that for maximum grain produced on Sandy's property, grain production per hectare needs to be about 59% more than that given in part (i) above. 2

THE END

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Q1

a) $\lim_{x \rightarrow 0} \frac{\sin 3x}{x} = 3 \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} = 3$

b) B is (1, 3)

or

$(-2, 5) \rightarrow (x, y)$
 $-3:2$

$\frac{-4-3x}{-3+2} = 7$

$\frac{10-3y}{-3+2} = -1$

$-4-3x = -7$

$10-3y = 1$

$-3x = -3$

$-3y = -9$

$x = 1$

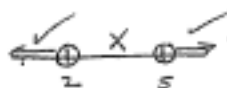
$y = 3$

c) $\frac{x+1}{x-2} = 2$

$x \neq 2$

$x+1 = 2x-4$

$x = 5$



$x > 5$ or $x < 2$

d) $y = \tan^{-1} 2x$

$\frac{dy}{dx} = \frac{1}{1+(2x)^2}$

$= \frac{1}{1+4x^2}$

when $x = \frac{1}{2}$

$m_{\text{tang}} = \frac{1}{1+4 \cdot \frac{1}{4}}$

$= \frac{1}{2}$

①

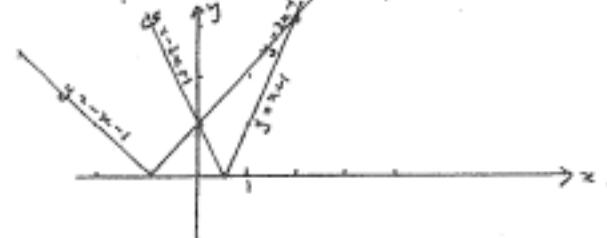
②

②

Q1

e) $\int_0^3 \frac{1}{\sqrt{9-x^2}} dx = \left[\sin^{-1} \frac{x}{3} \right]_0^3$
 $= \sin^{-1} \frac{3}{3} - \sin^{-1} 0$
 $= \frac{\pi}{2}$

f) $y = |2x-1|$ and $y = |x+1|$



$2x-1 = x+1$

$x = 2$

$-2x+1 = x+1$

$-3x = 0$

$x = 0$

$\therefore |2x-1| \leq |x+1|$ when $0 \leq x \leq 2$.

Q2.

$$\frac{\sin 2\theta}{\sin \theta} - \sec \theta = \frac{\cos 2\theta}{\cos \theta}$$

$$\begin{aligned} \text{a) LHS} &= \frac{2 \sin \theta \cos \theta}{\sin \theta} - \frac{1}{\cos \theta} \\ &= 2 \cos \theta - \frac{1}{\cos \theta} \\ &= \frac{2 \cos^2 \theta - 1}{\cos \theta} \\ &= \frac{\cos 2\theta}{\cos \theta} \end{aligned}$$

$$\begin{aligned} \text{b) } u &= 2t-1 && \text{when } t=1 \quad u=1 \\ \frac{du}{dt} &= 2 && \text{when } t=\frac{1}{2} \quad u=0. \end{aligned}$$

$$\begin{aligned} \int_{\frac{1}{2}}^1 4t(2t-1) dt &= \int_0^1 2(u+1)u^{\frac{1}{2}} du \\ &= \int_0^1 u^{\frac{3}{2}} + u^{\frac{1}{2}} du \\ &= \left[\frac{u^{\frac{5}{2}}}{\frac{5}{2}} + \frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^1 \\ &= \frac{2}{5} + \frac{2}{3} \\ &= \frac{14}{15} \end{aligned}$$

$$\text{c) } \tan \theta = \left| \frac{3-m}{1+3m} \right|$$

$$1 = \frac{3-m}{1+3m}$$

$$1+3m = 3-m$$

$$4m = 2$$

$$m = \frac{1}{2}$$

$$-1 = \frac{3-m}{1+3m}$$

$$-1-3m = 3-m$$

$$-2m = 4$$

$$m = -2$$

$$\text{d) } \tan 2\theta - \cot \theta = 0 \quad 0 \leq \theta \leq \pi$$

(2)

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} - \frac{1}{\tan \theta} = 0$$

$$2 \tan^2 \theta = 1 - \tan^2 \theta$$

$$3 \tan^2 \theta = 1$$

$$\tan \theta = \pm \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\text{Check } \theta = \frac{\pi}{2} \quad 0 - 0 = 0 \text{ True}$$

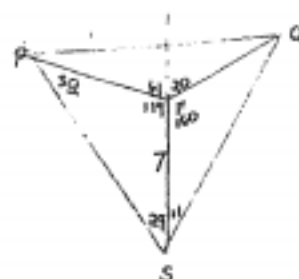
(4)

$$\therefore \text{Sols } \theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$$

(3)

Q4

c.



- (i) Consider
- $\triangle PFS$
- .

$$\frac{PF}{\sin 29} = \frac{7}{\sin 32}$$

$$PF = \sin 29 \times \frac{7}{\sin 32}$$

- (ii) Similarly considering
- $\triangle QFS$

$$QF = \sin 11 \times \frac{7}{\sin 9}$$

Now considering $\triangle PQF$

$$PQ^2 = PF^2 + QF^2 - 2 PF \times QF \cos 81^\circ$$

$$15. \quad d^2 = \frac{\sin^2 29 \times 49}{\sin^2 32} + \frac{\sin^2 11 \times 49}{\sin^2 9} - 2 \times \frac{\sin 29 \times 7}{\sin 32} \times \frac{\sin 11 \times 7}{\sin 9} \times \cos 81$$

$$= 49 \left[\frac{\sin^2 29}{\sin^2 32} + \frac{\sin^2 11}{\sin^2 9} - \frac{2 \sin 29 \sin 11 \cos 81}{\sin 32 \sin 9} \right]$$

Kate

Q4

$$x^3 + 2x^2 - 3x + 5 = 0$$

$$(i) \quad \alpha + \beta + \gamma = -\frac{b}{a} = -2$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = -3$$

$$(ii) \quad \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2\alpha\beta - 2\alpha\gamma - 2\beta\gamma$$

$$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= 4 - 2 \times -3$$

$$= 10$$

$$b) \quad P(x) = Q(x)(x+1) + ax + b$$

$$P(1) = -a + b = 5 \quad (1)$$

$$P\left(-\frac{1}{2}\right) = -\frac{1}{2}a + b = 3 \quad (2)$$

$$(1) - (2) \quad -\frac{1}{2}a = 2$$

$$a = -4$$

$$b = 1$$

$$\therefore \text{Remainder} = -4x + 1$$

Kate is the neatest writer

Kate

Q 3.

$$\begin{aligned}
 \text{(i)} \quad \lim_{n \rightarrow \infty} \left(\frac{1^3 + 2^3 + 3^3 + \dots + n^3}{n^4} \right) &= \lim_{n \rightarrow \infty} \frac{n^2(n+1)^2}{4n^4} \\
 &= \lim_{n \rightarrow \infty} \frac{n^4 + 2n^3 + n^2}{4n^4} \\
 &= \lim_{n \rightarrow \infty} \frac{1 + \frac{2}{n} + \frac{1}{n^2}}{4} \\
 &= \frac{1}{4}
 \end{aligned}$$

$$c. \text{ let } \sin x + \sqrt{3} \cos x = R \sin(x + \alpha)$$

$$R \cos \alpha = 1 \quad (1)$$

$$R \sin \alpha = \sqrt{3} \quad (2)$$

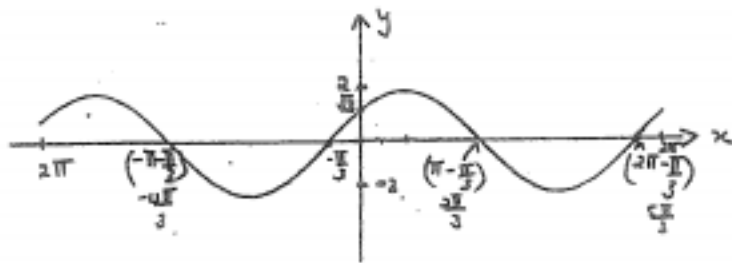
$$\textcircled{2} \quad \tan \alpha = \sqrt{3}$$

$$\textcircled{1} \quad \alpha = \frac{\pi}{3}$$

$$R^2 = 4$$

$$R = 2$$

$$\therefore \sin x + \sqrt{3} \cos x = 2 \sin\left(x + \frac{\pi}{3}\right)$$



$$\text{(ii)} \quad 2 \sin\left(x + \frac{\pi}{3}\right) = \sqrt{2}$$

$$\sin\left(x + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}}$$

$$x + \frac{\pi}{3} = n\pi + (-1)^n \frac{\pi}{4}$$

$$\therefore x = n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{3}$$

①

$$Q 3 \quad a) \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos^2\left(\frac{x}{2}\right) dx$$

②

$$\frac{1}{2} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (\cos x + 1) dx$$

$$= \frac{1}{2} \left[\sin x + x \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left[1 + \frac{\pi}{2} - \left(\frac{1}{2} + \frac{\pi}{6} \right) \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} + \frac{\pi}{3} \right]$$

$$= \frac{1}{4} + \frac{\pi}{6}$$

②

$$b) \text{ let } S_n = 4(1^3 + 2^3 + \dots + n^3)$$

③

$$\text{Required to prove } S_n = n^2(n+1)^2$$

$$\text{For } n=1 \quad \text{LHS} = 4(1^3) = 4$$

$$\text{RHS} = 1^2(1+1)^2 = 4$$

Statement is true for $n=1$.

②

Assume statement is true for $n=k$

$$S_k = 4(1^3 + 2^3 + \dots + k^3) = k^2(k+1)^2$$

when $n=k+1$

$$\begin{aligned}
 S_{k+1} &= 4(1^3 + 2^3 + \dots + k^3 + (k+1)^3) \\
 &= 4(1^3 + 2^3 + \dots + k^3) + 4(k+1)^3 \\
 &= k^2(k+1)^2 + 4(k+1)^3 \\
 &= (k+1)^2(k^2 + 4(k+1)) \\
 &= (k+1)^2(k+2)^2
 \end{aligned}$$

Thus if it is true for $n=k$ it is true for $n=k+1$
 It is true for $n=1$ & hence it is true for $n=2$ & so on

Q6.

$$\frac{d^2x}{dt^2} = -16x$$

$$\begin{aligned} (i) \quad x &= a \cos(4t + \alpha) \\ \dot{x} &= -4a \sin(4t + \alpha) \\ \ddot{x} &= -16a \cos(4t + \alpha) \\ &= -16x \end{aligned}$$

$\therefore x = a \cos(4t + \alpha)$ is a soln

(i) When $t=0$ $v=4$

$$4 = -4a \sin \alpha$$

$$\text{i.e. } -1 = a \sin \alpha \quad (1)$$

$$\text{When } t=0 \quad x=5$$

$$5 = a \cos \alpha \quad (2)$$

$$\text{i.e. } (1)^2 + (2)^2 \quad (\sin^2 \alpha + \cos^2 \alpha = 1)$$

$$a^2 = (-1)^2 + (5)^2$$

$$a^2 = 26$$

$$a = \sqrt{26}$$

OR

$$\text{using } v^2 = n^2(a^2 - x^2)$$

$$16 = 16(a^2 - 25)$$

$$1 = a^2 - 25$$

$$a^2 = 26$$

$$a = \sqrt{26}$$

(iii) Max speed occurs when $\sin(4t + \alpha) = 1$.

$$\ddot{x} = -4\sqrt{26} \sin(4t + \alpha)$$

$$= -4\sqrt{26}$$

$$\text{speed}_{\max} = |-4\sqrt{26}|$$

$$= 4\sqrt{26}$$

Q6

b $x^2 = 4ay$

$$y = \frac{1}{4a} x^2$$

$$(i) \frac{dy}{dx} = \frac{1}{2a} x$$

$$m_{\text{tangent}} = \frac{1}{2a} \times 2ap$$

$$= p$$

Eqⁿ of tangent at P

$$y - ap^2 = p(x - 2ap^2)$$

$$y = px - ap^2 \quad (1)$$

$$(ii) y = qx - aq^2 \quad (2)$$

(iii) Solving eqns for tangents simult.

$$p = (p - q)x - a(p^2 - q^2)$$

$$x = \frac{a(p^2 - q^2)}{p - q} = a \frac{(p+q)(p-q)}{p-q}$$

$$x = a(p+q)$$

$$(i) y = ap(p+q) - ap^2$$

$$= apq$$

$$\therefore T \text{ is } (a(p+q), apq)$$

(iv) T lies on $x^2 = 4ay$

$$\text{i.e. } a^2(p+q)^2 = -4a^2pq$$

$$(p+q)^2 = -4pq$$

$$p^2 + q^2 + 2pq = -4pq$$

$$p^2 + q^2 = -6pq$$

(v) M is $\left(\frac{2a(p+q)}{2}, \frac{a(p^2+q^2)}{2}\right)$

$$= \left(a(p+q), \frac{a(p^2+q^2)}{2}\right)$$

Q 5

$$f(x) = \frac{x-1}{x^2}$$

$$\begin{aligned} \text{U) } f'(x) &= \frac{x^2 \cdot 1 - 2x(x-1)}{x^4} \\ &= \frac{x^2 - 2x^2 + 2x}{x^4} \\ &= \frac{x(2-x)}{x^4} \\ &= \frac{2-x}{x^3} \end{aligned}$$

$$f'(x) = 0 \text{ when } x=2.$$

$$\begin{aligned} f''(x) &= \frac{x^3 \cdot (-1) - 3x^2(2-x)}{x^6} \\ &= \frac{-x^3 - 6x^2 + 3x^3}{x^6} \\ &= \frac{2x^2(x-3)}{x^6} \\ &= \frac{2(x-3)}{x^4} \end{aligned}$$

$$\text{when } x=2, f''(x) < 0$$

$\therefore$ Only stationary pt $(2, \frac{1}{4})$ which is a max

$$\text{U) } f''(x) = \frac{2(x-3)}{x^4} = 0 \text{ when } x=3$$

x	3^-	3	3^+
$f''(x)$	$-$	0	$+$

Change in concavity

$\therefore$ there is a pt of inflexion at $(3, \frac{2}{9})$

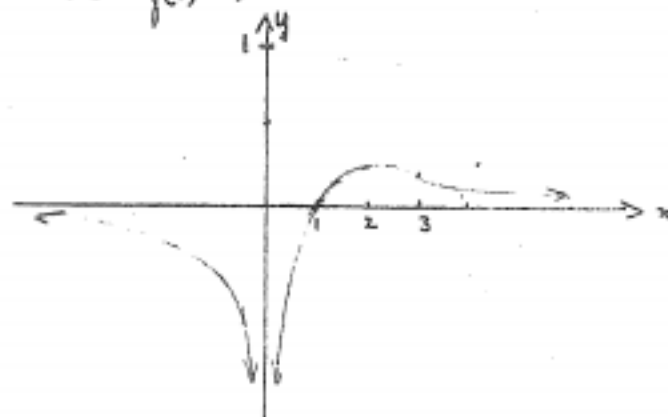
(3)

Q 5,

$$\text{(iii) As } x \rightarrow \infty, f(x) \rightarrow 0$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow 0$$

$$\text{(iv) As } x \rightarrow 0, f(x) \rightarrow -\infty$$



$$\begin{aligned} \text{b) } \frac{d}{dx} \left(\frac{1}{2} v^2 \right) &= \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \frac{dv}{dx} \\ &= v \cdot \frac{dv}{dx} \\ &= \frac{dx}{dt} \cdot \frac{dv}{dx} \\ &= \frac{dv}{dt} \\ &= \ddot{x} \end{aligned}$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 8x - 3x^2$$

$$\frac{1}{2} v^2 = \int 8x - 3x^2 dx$$

$$\frac{1}{2} v^2 = 4x^2 - x^3 + C$$

$$\text{when } x=0, v=4 \therefore C=8$$

$$v^2 = 8x^2 - 2x^3 + 16$$

$$\text{when } x=1, v^2 = 8 - 2 + 16$$

(1)

(1)

(2)

(2)

(2)

Q6

For M

$$x = a(p+q)$$

$$x^2 = a^2(p^2 + q^2 + 2pq)$$

$$= a^2(-6pq + 2pq)$$

$$x^2 = -4a^2pq \quad (1)$$

$$y = \frac{a}{2}(p^2 + q^2)$$

$$= \frac{a}{2}x - 6pq$$

$$y = -3apq$$

$$\therefore \frac{y}{-3a} = pq$$

Sub in (1)

$$x^2 = -4a^2 \times \frac{y}{-3a}$$

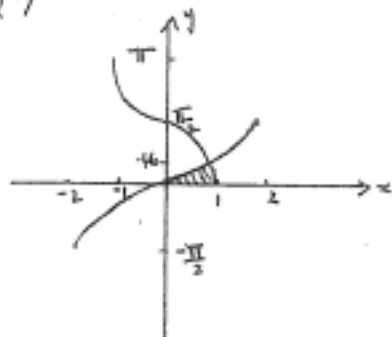
$$x^2 = \frac{4a}{3}y$$

$$3x^2 = 4ay$$

(2)

Q 7

(i)



(2)

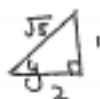
or ---

$$(ii) y = \cos^{-1} x$$

when $x = \frac{2}{\sqrt{5}}$

$$y = \cos^{-1} \frac{2}{\sqrt{5}}$$

i.e. $\cos y = \frac{2}{\sqrt{5}}$



$$\text{Also } \sin y = \frac{1}{\sqrt{5}}$$

$$\text{Consider } y = \sin^{-1} x$$

$$y = \sin^{-1} \left(\frac{2}{\sqrt{5}} \right)$$

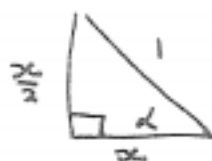
$$\text{i.e. } \sin y = \frac{1}{\sqrt{5}} \text{ True}$$

$$\therefore \text{Curves intersect at } x = \frac{2}{\sqrt{5}}$$

$$(iii) y = \sin^{-1} \frac{x}{2}$$

Inv. fn is $\frac{x}{2}$ $\sin y = \frac{x}{2}$

i.e. $x = 2 \sin y$



$$x^2 + \frac{x^2}{4} = 1$$

$$5x^2 = 4$$

$$x = \pm \frac{2}{\sqrt{5}}$$

$$x = \frac{2}{\sqrt{5}}$$

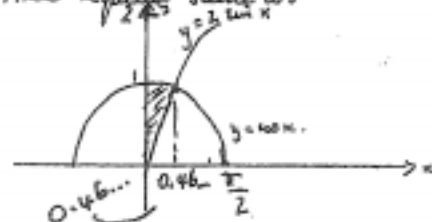
but $x > 0$ from graph

Q 7.

$$(iv) y = \cos^{-1} x$$

$$\text{Inv } x = \cos y.$$

Area required same as



$$\text{Area} = \int_0^{0.46...} (\cos x - 2 \sin x) dx$$

$$\left[\sin x + 2 \cos x \right]_0^{0.46...}$$

$$[0.443 + 1.792 - 0 - 2]$$

$$= 0.24 \text{ units}^2$$

7 b
(i) To maximise hectares irrigated,
we need to minimise water per hectare. (w)

$$\frac{dW}{dg} = 2Cg - \frac{D}{g^2} \quad \left. \vphantom{\frac{dW}{dg}} \right\} \frac{1}{2}$$

$$= 0 \text{ for stationary pts.} \quad \left. \vphantom{= 0} \right\} \frac{1}{2}$$

$$\text{ie } g^3 = \frac{D}{2C}$$

$$g = \left(\frac{D}{2C}\right)^{\frac{1}{3}} \quad (1)$$

$$\frac{d^2W}{dg^2} = 2C + \frac{2D}{g^3} \quad \left. \vphantom{\frac{d^2W}{dg^2}} \right\} \frac{1}{2}$$

$$> 0 \text{ since } C, D, g > 0 \text{ min. TP.} \quad \left. \vphantom{> 0} \right\} \frac{1}{2}$$

(ii) Let G = tonnes of grain per kl water
We need to maximise G

$$G = \text{tonnes of grain per hectare} \times \text{hectares per kl water} \quad \left. \vphantom{G} \right\} \frac{1}{2}$$

$$= g \times \frac{1}{Cg^2 + \frac{D}{g}}$$

$$\frac{dG}{dg} = \frac{Cg^2 + \frac{D}{g} - g(2Cg - \frac{D}{g^2})}{(Cg^2 + \frac{D}{g})^2} \quad \left. \vphantom{\frac{dG}{dg}} \right\} \frac{1}{2}$$

= 0 for stationary pts

$$\text{ie } g^3 = \frac{2D}{C}$$

$$g = \left(\frac{2D}{C}\right)^{\frac{1}{3}} \quad (2)$$

$$\frac{dG}{dg} = -\frac{1}{g} (Cg^3 - 2D) \times \frac{1}{(Cg^2 + \frac{D}{g})^2}$$

g	$\sqrt[3]{\frac{2D}{C}} - \epsilon$	$\sqrt[3]{\frac{2D}{C}}$	$\sqrt[3]{\frac{2D}{C}} + \epsilon$
$\frac{dG}{dg}$	neg	0	pos
	neg	0	pos

$\therefore$ Max. T.R

Now from (1) & (2) above
comparing results

$$\sqrt[3]{\frac{2D}{C}} = \sqrt[3]{4}$$

$$\sqrt[3]{\frac{D}{2C}} = 1.587 \dots$$

$\approx 59\%$ more than before